

BIOS 794: Generalized Additive Models

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Introduction

- ▶ So far, the regression methods that we've used have assumed some linear function $\mathbf{X}'\beta$.
- ▶ In this section, we explore non-linear functions $h_m(\mathbf{X}) : \mathbb{R}^p \rightarrow \mathbb{R}$ for $m = 1, \dots, M$ and use the model

$$f(\mathbf{X}) = \sum_{m=1}^M \beta_m h_m(\mathbf{X})$$

- ▶ The $h_m(\mathbf{X})$ we'll consider are *basis expansions*.
- ▶ There are many types of **basis expansions**: *polynomials*, *piecewise-polynomials*, *splines*, and *wavelets* are a few.

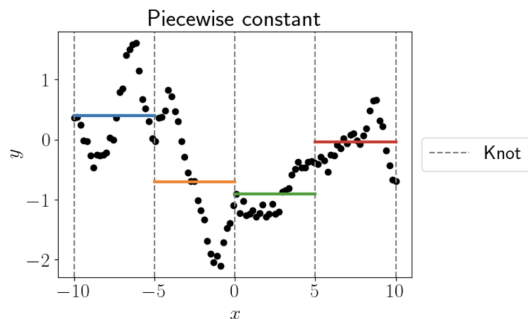
Piecewise-polynomials and splines

- ▶ For the moment, we'll consider one-dimensional $\mathbf{X}$, e.g., we don't know the impact of BMI and 6-month change in HbA1C.
- ▶ Piecewise constant with 2 knots

$$h_1(X) = I(X < \xi_1)$$

$$h_2(X) = I(\xi_1 \leq X < \xi_2)$$

$$h_3(X) = I(X \geq \xi_2)$$



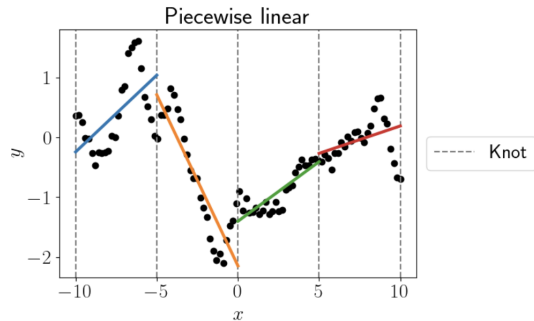
Piecewise-polynomials and splines

- Piecewise linear with 2 knots would have everything from the piecewise constant plus:

$$h_4(X) = X \times I(X < \xi_1)$$

$$h_5(X) = X \times I(\xi_1 \leq X < \xi_2)$$

$$h_6(X) = X \times I(X > \xi_2)$$



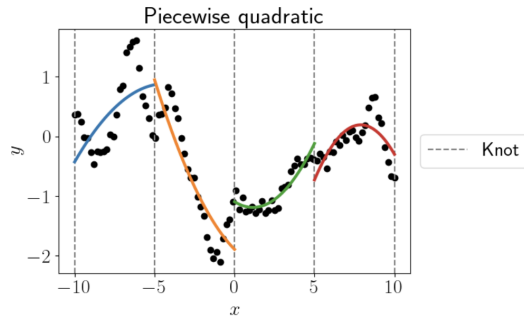
Piecewise-polynomials and splines

- Piecewise quadratic with 2 knots would have everything from the piecewise linear plus:

$$h_7(X) = X^2 I(X < \xi_1)$$

$$h_8(X) = X^2 I(\xi_1 \leq X < \xi_2)$$

$$h_9(X) = X^2 I(X > \xi_2)$$

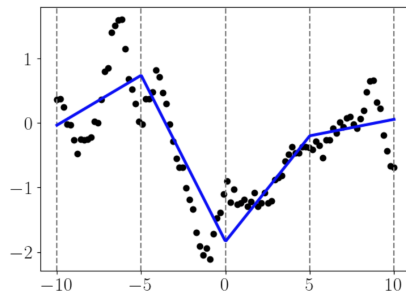


Piecewise-polynomials and splines

- ▶ The previous versions have discontinuities.
- ▶ Continuous piecewise linear version

$$h_1(X) = 1 \quad h_2(X) = X \quad h_3(X) = (X - \xi_1)_+$$

- ▶ Note that the first derivative (i.e., the slope) is not continuous.



Piecewise Cubic Polynomials

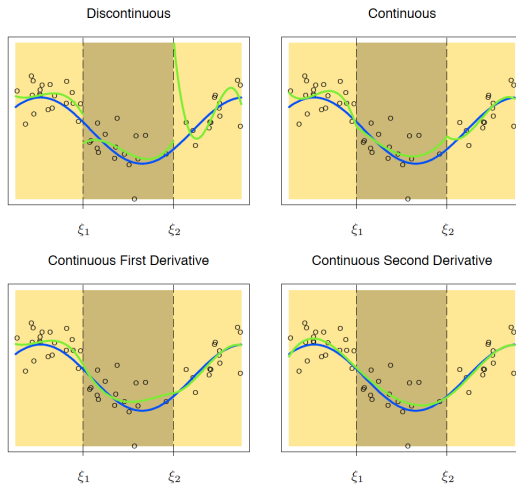


Figure: From ESL (online version).

Cubic splines

- ▶ Having a continuous second derivative leads to *cubic spline*
 - ▶ (note: continuous second derivative functions are smooth to the naked eye, going beyond second doesn't make them look more smooth)
- ▶ For our example we have

$$h_1(X) = 1 \quad h_3(X) = X^2 \quad h_5(X) = (X - \xi_1)_+^3$$

$$h_2(X) = X \quad h_4(X) = X^3 \quad h_6(X) = (X - \xi_2)_+^3$$

- ▶ The number of parameters is: $(3 \text{ regions}) \times (4 \text{ parameters per region}) - (2 \text{ knots}) \times (3 \text{ constraints per knot}) = 6$.

order- M splines

- ▶ An order- M spline (or a *piecewise-polynomial of order M*) has continuous derivatives to up to order $M - 2$ (for cubic spline $M = 4$).
- ▶ With knots at ξ_j for $j = 1, \dots, K$ we have

$$\begin{aligned}h_j(X) &= X^{j-1} \quad \text{for } j = 1, \dots, M \\h_{M+\ell}(X) &= (X - \xi_\ell)_+^{M-1}, \quad \text{for } \ell = 1, \dots, K\end{aligned}$$

- ▶ These **fixed knot splines** are known as *regression splines*

Natural Cubic Splines

- ▶ Polynomials are erratic towards the boundary of the data.
- ▶ This can be exacerbated with piecewise-polynomial splines.
- ▶ *Natural cubic splines* add the constraint that the function is linear beyond the boundary of the data.
- ▶ This constraint decreases the *degrees of freedom* of the model (i.e., we could add more interior knots) and adds stability to the model at and beyond the boundary.
 - ▶ it also may add bias, but some bias would be preferred since we don't have much data in this region.

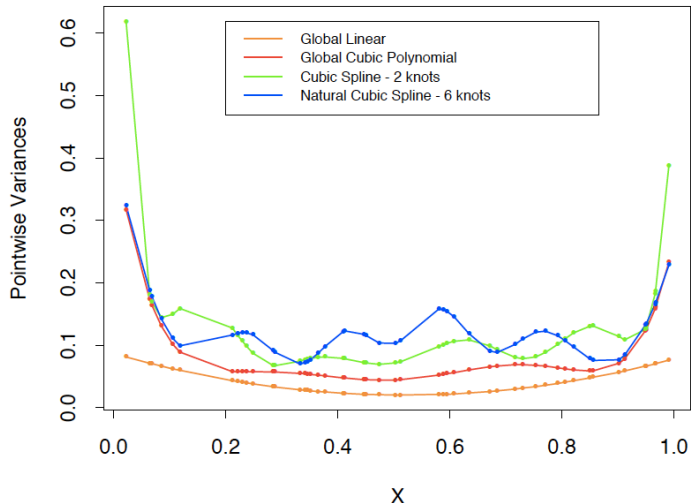


Figure: Pointwise variance of different spline methods. From ESL (online version) page 145.

Natural Cubic Splines

- ▶ Natural Cubic Splines with K knots are represented by K basis functions.
- ▶ They can be derived by reducing the basis of cubic spline by imposing the boundary constraints.
- ▶ In this case they take the form

$$N_1(X) = 1, \quad N_2(X) = X, \quad N_{k+2}(X) = d_k(X) - d_{K-1}(X),$$

where

$$d_k(X) = \frac{(X - \xi_k)_+^3 - (X - \xi_K)_+^3}{\xi_K - \xi_k}.$$

- ▶ Note that each of these basis functions has zero 2nd and 3rd derivative outside the boundary knots

Cubic B-Splines

- ▶ The previous splines we've discussed are good to learn from since they have relatively simple forms.
- ▶ In practice, (by far) the most used spline is the B-spline.
- ▶ Let $B_{i,m}(x)$ be the i th B-spline basis function of order m with knot sequence τ .
- ▶ The B-splines are defined recursively as $B_{i,1} = I(\tau_i \leq x < \tau_{i+1})$ for $i = 1, \dots, K + 2M - 1$ and

$$B_{i,m}(x) = \frac{x - \tau_i}{\tau_{i+m-1} - \tau_i} B_{i,m-1}(x) + \frac{\tau_{i+m} - x}{\tau_{i+m} - \tau_{i+1}} B_{i+1,m-1}(x)$$

- ▶ The most common is the $m = 4$ cubic B-splines. There are also natural cubic B-splines.

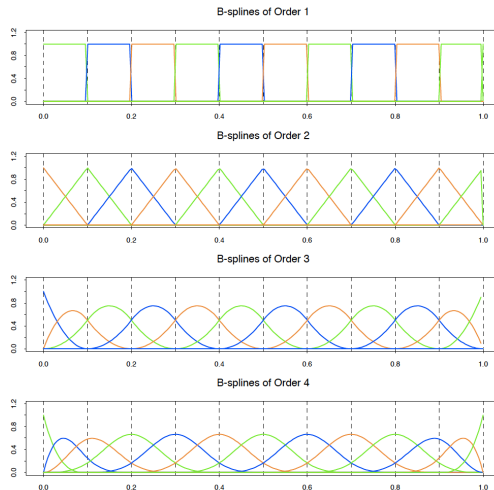


Figure: B-splines of varying order. From ESL (online version) page 188.

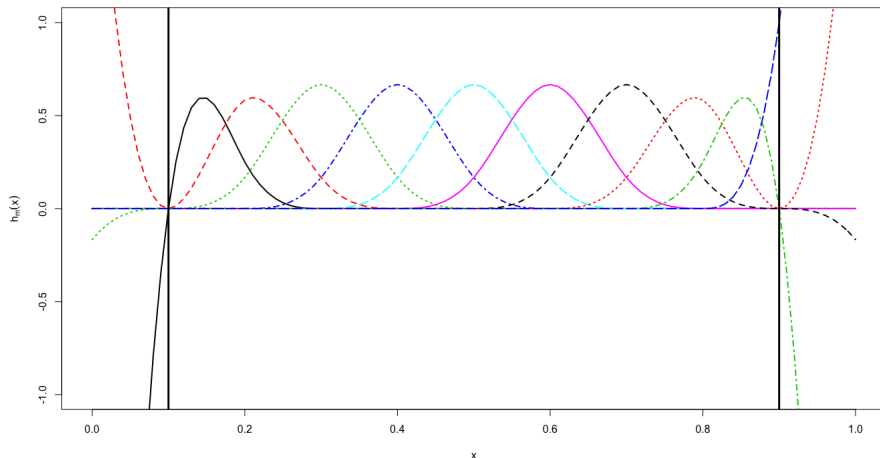


Figure: Cubic B-splines with $df = 10$ and boundary knots at 0.1 and 0.9.

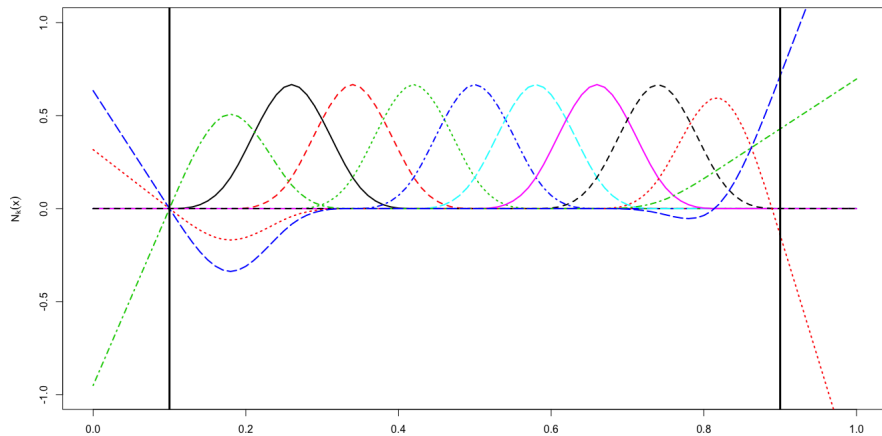


Figure: Natural cubic B-splines with $df = 10$ and boundary knots at 0.1 and 0.9.

Introduction

- ▶ The previous slides we had to choose the number and location of the interior knots of the spline function.
- ▶ To avoid the knot selection issue we can use the penalized RSS

$$RSS(f, \lambda) = \sum_{i=1}^N \{y_i - f(x_i)\}^2 + \lambda \int \{f''(t)\}^2 dt$$

where λ is the *smoothing parameter*

- ▶ Some special cases:
 - ▶ $\lambda = 0$: f interpolates the data
 - ▶ $\lambda = \infty$: reverts to the LS fit wrt x with no second derivative

Smoothing Spline Solution

- ▶ Interestingly, it can be shown that $RSS(f, \lambda)$ has an explicit, finite-dimensional, unique minimizer which is a *natural cubic spline* with knots at **the unique values of the x_i for $i = 1, \dots, N$** .
- ▶ While having (potentially) N knots might seem crazy the penalty term will shrink the spline coefficients to zero.
- ▶ We write the solution to this model as

$$f(x) = \sum_{i=1}^N N_j(x) \theta_j$$

where $N_j(x)$ represent the basis functions of this family of natural splines.

Smoothing Spline Solution

- ▶ We can re-write the penalized RSS as

$$RSS(f, \lambda) = (\mathbf{y} - \mathbf{N}\theta)'(\mathbf{y} - \mathbf{N}\theta) + \lambda\theta'\Omega_N\theta$$

where $\mathbf{N}_{ij} = N_j(x_i)$ and $\{\Omega_N\}_{jk} = \int N_j''(t)N_k''(t)dt$.

- ▶ Note the similarities of this RSS and ridge regression. In fact, this is known as a *generalized ridge regression*
- ▶ The solution is given by

$$\hat{\theta} = (\mathbf{N}'\mathbf{N} + \lambda\Omega_N)^{-1}\mathbf{N}'\mathbf{y}$$

Selecting λ via degrees of freedom

- ▶ We can estimate the *effective degrees of freedom* via

$$df_{\lambda} = \text{trace}(\mathbf{S}_{\lambda})$$

where $\mathbf{S}_{\lambda}$ is known as the smoother matrix.

- ▶ This gives us a way to specify the amount of smoothing in terms of the degrees of freedom (i.e., number of parameters).
- ▶ For example, we can set λ by solving $df_{\lambda} = 12$ for λ .
- ▶ The package that we'll use selects λ using REML.

Introduction

- ▶ So far we have focused on one-dimensional spline models. Each of the approaches have multidimensional analogs.
- ▶ Suppose $X \in \mathbb{R}^2$ and we have univariate function $h_{1j}(X_1)$ for $j = 1, \dots, M_1$ and $h_{2j}(X_2)$ for $j = 1, \dots, M_2$.
- ▶ Then the $M_1 \times M_2$ dimensional tensor product basis defined by

$$g_{jk}(X) = h_{1j}(X_1)h_{2k}(X_2) \quad j = 1, \dots, M_1 \quad k = 1, \dots, M_2$$

gives the two-dimensional function

$$g(X) = \sum_{j=1}^{M_1} \sum_{k=1}^{M_2} \theta_{jk} g_{jk}(X).$$

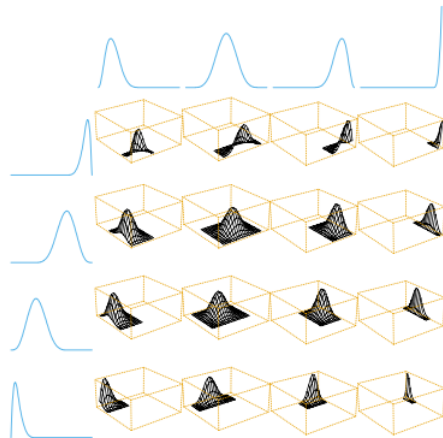


FIGURE 5.10. A tensor product basis of B-splines, showing some selected pairs. Each two-dimensional function is the tensor product of the corresponding one dimensional marginals.

Figure: A tensor product of basis of B-splines. From ESL (online version) page 163.

Penalization

- ▶ The above can be generalized to d dimensions, but the number of parameters grows exponentially (curse of dimensionality).
- ▶ Tensor basis functions can again be fitted with a penalized RSS

$$\sum_{i=1}^N \{y_i - f(x_i)\}^2 + \lambda J[f]$$

where J is a penalty functional for the f , which is no longer univariate.

Basis Functions in Practice: Why Use Them?

- ▶ **Check Relationships:** I often use basis functions to see whether the relationship between a predictor and outcome is really linear.
- ▶ **Capture Curves:** Expansions (like polynomials or splines) let us see curved or more complex patterns that a straight line would miss.
- ▶ **Flexible but Simple:** They still use the linear regression framework, so fitting and interpreting them is familiar.
- ▶ **Better Predictions:** If a predictor has a non-linear effect, including basis functions can improve model fit and prediction.

Basis Functions in Practice: Things to Watch For

- ▶ **Overfitting:** Adding too many terms can chase noise instead of the real signal.
- ▶ **Interpretability:** Higher-order terms or spline bases can make interpretation harder.
- ▶ **Correlation Among Terms:** Polynomial terms can be highly correlated with each other.
- ▶ **Choices Matter:** You must decide how many terms to add (e.g., polynomial degree, number of spline knots).
- ▶ **Edge Behavior:** Some basis expansions behave strangely at the extremes of the data.